

Quality

- The totality of features and characteristics of a product or service that bears on its ability to satisfy given needs.”
- Quality is meeting or exceeding customer expectations.

Customer Quality Measures:

1. Feature-Based Measures:

Feature-based measures assess the presence or absence of specific attributes or functionalities that customers value in a product or service. These measures focus on the characteristics that contribute to the utility, convenience, or desirability of the offering. Some common feature-based measures include:

Feature Completeness: Evaluates whether the product or service includes all the features that customers expect or desire.

Innovation: Assesses the degree of novelty or uniqueness in the product's features compared to competitors.

Customization Options: Measures the availability and flexibility of customization features to meet diverse customer preferences.

Compatibility: Determines whether the product is compatible with other systems, devices

User-Friendliness: Evaluates the ease of use, intuitiveness, and accessibility of features, interfaces, and controls.

Safety Features: Assesses the presence of safety mechanisms or features that protect users from harm or accidents.

2. Performance Measures:

Performance measures focus on how well a product or service performs in terms of meeting customer needs, expectations, or standards.

Reliability: Measures the consistency and dependability of the product or service .

Speed: Assesses the responsiveness or processing time of the product or service.

Accuracy: Determines the precision or correctness of outputs.

Efficiency: Measures the resource utilization or cost-effectiveness of the product or service in achieving desired outcomes.

Availability: Evaluates the accessibility and uptime of the product or service, ensuring it is readily accessible to users when needed.

Scalability: Assesses the ability of the product or service to accommodate increasing demands or usage levels without significant performance degradation.

Consistency: Measures the uniformity or stability of performance across different usage scenarios, environments, or user groups.

Different Views of Quality

Customer Views of Quality:

Fitness for Use: This perspective focuses on whether a product or service meets the customer's needs and expectations. It considers whether the product or service can perform its intended function effectively.

Performance: Customers often assess quality based on the performance of a product or service. This includes factors such as reliability, durability, speed, accuracy, and efficiency.

Reliability: Customers value consistency and dependability. They want products and services that perform consistently over time without failure or defects.

Durability: Durability refers to the ability of a product to withstand wear, tear, and deterioration over time. Customers expect products to have a reasonable lifespan without significant degradation in performance.

Features: Customers may perceive higher quality in products that offer additional features or functionalities beyond basic requirements. These extra features enhance the utility and value of the product.

Aesthetics: The appearance, design, and sensory appeal of a product contribute to its perceived quality. Customers often associate visually appealing or well-designed products with higher quality.

Serviceability: For services, quality may be judged based on factors such as responsiveness, professionalism, ease of access, and the ability to resolve issues promptly.

Producer View of Quality:

Conformance to Specifications: From the producer's perspective, quality often involves meeting predefined standards and specifications.

Cost of Quality: Producers consider the cost implications of quality-related activities, including **prevention costs** (e.g., training, quality control),

appraisal costs (e.g., inspection, testing),

scrap and rework costs: internal costs of defective products.

warranty costs: external costs for product failures in the field

Balancing these costs is essential for achieving optimal quality outcomes.

Process Efficiency: Producers focus on optimizing production processes to ensure consistent quality while minimizing costs and waste.

Continuous Improvement: Producers strive for continuous improvement in quality through initiatives such as Total Quality Management (TQM), Six Sigma, or Lean Manufacturing.

Customer Satisfaction and Loyalty: Producers recognize the importance of customer satisfaction and loyalty in maintaining a competitive edge. By delivering high-quality products and services, producers can enhance customer satisfaction, build brand reputation, and foster long-term customer relationships.

Market Positioning: Quality plays a crucial role in shaping a company's reputation and market positioning.

TQM

TQM is a management philosophy which seeks to integrate all organizational functions (marketing, finance, design, engineering, production, customer service ...) to focus on meeting customer needs and organizational objectives.

Some Basic Tenets of TQM

1. The customer determines quality.
2. Improving quality requires the establishment of effective quality metrics. We must speak with data not just opinions.
3. People working within systems create quality through teamwork.
4. Quality is a moving target. It requires a commitment toward sustained (strategic and systematic) continuous improvement.
5. Prevention, not detection, is the key to producing high quality. We must design quality into processes and reduce variability.
6. Top Management must provide leadership and support for all (total) quality initiatives.

Productivity

Productivity refers to the efficiency with which resources are utilized to achieve desired outcomes or produce goods and services.

Productivity is the ratio of output to input.

labor productivity

Production per worker or per hour worked is referred to as labor productivity. Workforce skills, technological development, managerial strategies, and adjustments to other inputs are all factors that may affect labor productivity.

Capital Productivity:

This measure assesses the efficiency of capital investment by calculating the amount of output generated per unit of capital input.

Machine Productivity:

This measure evaluates the productivity of machines or equipment by calculating the amount of output produced per hour of machine operation

Energy Productivity:

This measure assesses the efficiency of energy usage by calculating the amount of output produced per kilowatt-hour of energy consumed.